

EuroBERT: Scaling Multilingual Encoders for European Languages



Nicolas Boizard^{1,3} Hippolyte Gisserot-Boukhlef^{2,3} Duarte M. Alves^{4,5}

André Martins^{† 4,5,6} Ayoub Hammal^{† 7,8,9} Caio Corro^{† 8,10,11} Céline Hudelot^{† 3}
 Emmanuel Malherbe^{† 2} Etienne Malaboeuf^{† 12} Fanny Jourdan^{† 13} Gabriel
 Hauteux^{† 12} João Alves^{† 6} Kevin El-Haddad^{† 1,17} Manuel Faysse^{† 3,14} Maxime
 Peyrard^{† 8,15} Nuno M. Guerreiro^{† 3,4,5,6} Patrick Fernandes^{† 4,5,18} Ricardo Rei^{† 6}
 Pierre Colombo^{* 3,16}

¹Diabolocom, ²Artefact Research Center, ³MICS, CentraleSupélec, Université Paris-Saclay, ⁴Instituto Superior Técnico & Universidade de Lisboa (Lisbon ELLIS Unit), ⁵Instituto de Telecomunicações, ⁶Unbabel, ⁷Université Paris-Saclay, ⁸CNRS, ⁹LISN, ¹⁰INSA Rennes, ¹¹IRISA, ¹²CINES, ¹³IRT Saint Exupéry, ¹⁴Illuin Technology, ¹⁵Université Grenoble Alpes, Grenoble INP, LIG, ¹⁶Equall, ¹⁷ISIA Lab, ¹⁸Carnegie Mellon University

Equal contribution, [†]Ordered alphabetically by the first name, ^{*}Senior advisor

General-purpose multilingual vector representations, used in retrieval, regression, and classification, are traditionally obtained from bidirectional encoder models. Despite their wide applicability, encoders have been recently overshadowed by advances in generative decoder-only models. However, many innovations driving this progress are not inherently tied to decoders. In this paper, we revisit the development of multilingual encoders through the lens of these advances, and introduce EuroBERT, a family of multilingual encoders covering European and widely spoken global languages. Our models outperform existing alternatives across a diverse range of tasks, spanning multilingual capabilities, mathematics, and coding, and natively support sequences of up to 8,192 tokens. We also examine the design decisions behind EuroBERT, offering insights into our dataset composition and training pipeline. We publicly release the EuroBERT models, including intermediate training checkpoints, together with our training framework.

Contact: firstname.lastname@centralesupelec.com, duartemalves@tecnico.ulisboa.pt

Website: <https://huggingface.co/EuroBERT>

Date: March 10, 2025

1 Introduction

Many important tasks in Natural Language Processing (NLP), including information retrieval, classification, or regression, are built upon general-purpose vector representations. These representations are traditionally obtained from *bidirectional* encoder models, which aggregate information from the left and right contexts of each token (Devlin et al., 2019; Conneau et al., 2020; He et al., 2023). In contrast, recent advances in generative modeling have shifted the research community’s attention towards *unidirectional* architectures (Bai et al., 2023; Llama Team, 2024; OLMo et al., 2025). Notably, these efforts have identified several key performance drivers that span architectural advances, data improvements, and increased scale. Yet, despite no apparent barrier to transferring these insights to *bidirectional* architectures, little effort has been devoted towards this objective, forcing practitioners to depend on outdated models.

In this paper, we introduce a refreshed recipe for training general-purpose multilingual encoders, resulting in the EuroBERT family. Drawing inspiration from recent progress in decoder models, our models feature an updated architecture (§2.1), and are trained on a 5T-token multilingual dataset, covering widely spoken European and global languages,

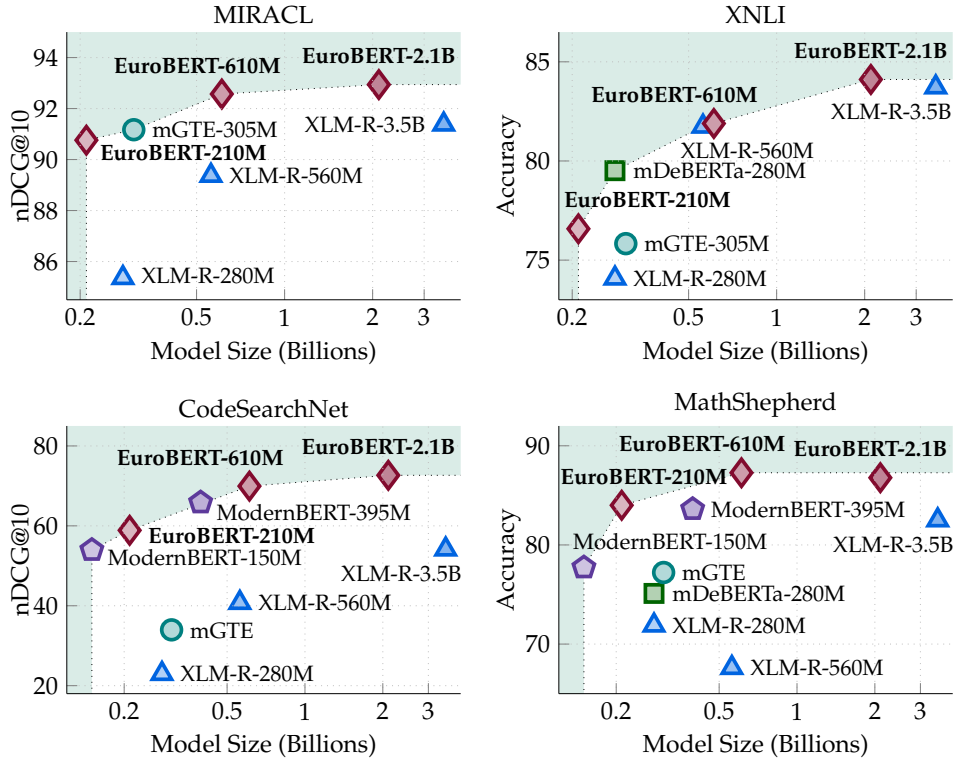


Figure 1: Pareto plots for multilingual tasks (top), showing retrieval performance on MIRACL and sentence classification on XNLI, and for math and code tasks (bottom), featuring CodeSearchNet and MathShepherd. The shaded regions indicate the Pareto frontiers.

along with mathematics and code (§2.2). We adopt a masked language modeling objective, and employ a two-phase training pipeline, adjusting the data distribution in the second training phase to improve downstream performance (§2.3).

We extensively evaluate the EuroBERT models, comparing with similarly sized alternatives across a suite of tasks representative of real-world encoder applications (§3). Our models match or exceed the performance of alternative models, such as XLM-RoBERTa (Conneau et al., 2020) and mGTE-MLM (Zhang et al., 2024), on multilingual retrieval, classification and regression tasks, and outperform them on code and mathematics tasks (Figure 1).

We also examine the impact of our design choices through systematic ablations on several components of our annealing recipe (§4). We explore the choice of masking ratio, showing that while higher masking ratios benefit retrieval tasks, lower ratios improve sentence classification. Additionally, we highlight that including data for code and mathematics improves multilingual retrieval, but degrades classification accuracy. Contrary to expectations, we also observe that only selecting documents with high educational value degrades performance, and that instead encoders benefit from a broader range of data sources.

Accompanying this work, we release the EuroBERT family, comprising three models with 210M, 610M and 2.1B parameters. To facilitate future research, we also release intermediate training checkpoints, as well as our training framework.

2 EuroBERT: A Refreshed Multilingual Encoder

The EuroBERT models are available in three sizes (210M, 610M, and 2.1B parameters) and closely follow the Llama 3 architecture (Llama Team, 2024) (§2.1). They are trained on a large multilingual corpus, which also includes code and mathematics (§2.2). Their training

pipeline has two stages, pre-training and annealing, and employs the masked language modeling (MLM) objective (§2.3).

2.1 Architecture

The EuroBERT models are based on a standard dense transformer (Vaswani et al., 2017), with several architectural changes. Similarly to Llama 2 (Touvron et al., 2023), we remove all biases. Additionally, we incorporate grouped query attention (Ainslie et al., 2023), swish gated linear units (Shazeer, 2020), root mean square layer normalization (Zhang & Sennrich, 2019), and rotary position embeddings (Su et al., 2024).¹

2.2 Dataset

To train EuroBERT, we construct a multilingual 5T-token corpus — 4.8T tokens for pre-training and 200B for annealing — which includes 15 languages: English, French, German, Spanish, Chinese, Italian, Russian, Polish, Portuguese, Japanese, Vietnamese, Dutch, Arabic, Turkish, and Hindi.² Following prior work on curriculum learning (Hu et al., 2024), we adjust the data distribution to emphasize higher-quality datasets during annealing.

Pre-training mixture. We use FineWeb (Penedo et al., 2024) for English, and CulturaX (Nguyen et al., 2024) for multilingual data. We also incorporate EuroLLM parallel data (Martins et al., 2024), which can improve cross-lingual transfer (Conneau & Lample, 2019; Reid & Artetxe, 2022; 2023), by concatenating to-English and from-English translation pairs, separated by a special `<|parallel_sep|>` token. Finally, inspired by the benefits of training on code for decoder models (Aryabumi et al., 2024), we add 38 programming languages from The Stack v2 (Lozhkov et al., 2024) and Proof-Pile-2 (Azerbayev et al., 2024), which we find improves multilingual retrieval (§4).

Annealing mixture. We classified data not seen during pre-training into four quality levels according to educational value using the EuroLLM classifier (Martins et al., 2024). We then kept the documents above the third threshold which, contrary to our expectations, improved performance on downstream tasks (§4). Additionally, we adjusted the data distribution based on multiple ablations (§4). Specifically, we decreased English while proportionally increasing the remaining languages. We also decreased the amount of code and math while increasing parallel data.³

2.3 Training Recipe

Masked language modeling. We pre-train EuroBERT models with a 50% masking ratio, following the insights from Wettig et al. (2023), which find that masking 15% and 30% of tokens is sub-optimal, and that larger models benefit from higher masking ratios. For the subsequent annealing, however, we lower the masking ratio to 10% based on downstream evaluations (§4), aligning with the findings from Yang et al. (2023) and Ankner et al. (2024).

Hyperparameters. We employed the Warmup-Stable-Decay (WSD) scheduler (Shen et al., 2024), with a linear warm-up phase of 2,000 steps, a constant learning rate of 1×10^{-4} during pre-training, and a cosine scheduler decaying to 0 during the annealing phase. During pre-training, we packed sentences to 2,048 tokens and used a Rotary Position Embedding (RoPE) value of 10,000. In the annealing phase, we increased the RoPE theta to 250,000 and randomly cropped our training documents to lengths between 12 and 8,192 tokens. We adopted this approach because, due to pre-processing constraints, our training data had already been segmented into fixed-length documents, making standard variable-length training infeasible. Therefore, we introduced random cropping of these fixed-length

¹We provide more architecture details in Appendix A.

²These languages were selected to balance European and widely spoken global languages, and ensure representation across diverse alphabets and language families.

³We provide further details on our pretraining and annealing datasets in Appendix C.

sequences as an approximation of variable-length training. Surprisingly, we found that this approach outperforms training only on fixed lengths (§4), further highlighting the necessity for variable length documents during long context training (Gao et al., 2024).

Infrastructure. We trained EuroBERT on Adastra, using 92 MI250X GPUs for EuroBERT-210M, 384 MI250X GPUs for EuroBERT-610M, and 96 MI300A GPUs for EuroBERT-2.1B, for a total of 200k GPU hours. Our training framework incorporates FlashAttention (Dao, 2023), fused cross-entropy from LigerKernel (Hsu et al., 2024), torch.compile (Ansel et al., 2024), and hybrid sharding with Fully Sharded Data Parallel (Zhao et al., 2023).

3 Evaluation

3.1 Evaluation Setup

Datasets and tasks. We select a suite of tasks to cover various real-world use cases for encoders. For multilingual tasks, we evaluate retrieval using MIRACL (Zhang et al., 2023), MLDR (Chen et al., 2024), WikipediaRetrieval⁴, and CC-News (de Gibert et al., 2024). We assess sentence classification with XNLI (Conneau et al., 2018), PAWS-X (Yang et al., 2019), AmazonReviews (Keung et al., 2020) and MassiveIntent (Keung et al., 2020). Additionally, we evaluate token classification using the NER task from the XGLUE benchmark (Liang et al., 2020). We evaluate sequence regression on the WMT quality estimation task (Bojar et al., 2017; 2018; Barrault et al., 2019; 2020; Akhbardeh et al., 2021; Kocmi et al., 2022), and on summary evaluation using SeaHorse (Clark et al., 2023). For code-related tasks, we evaluate retrieval on CodeSearchNet (Husain et al., 2019) and DupStackMath (Hoogeveen et al., 2015), and classification on CodeDefect (Zhou et al., 2019) and CodeComplexity (Jeon et al., 2023). Finally, in the mathematical domain, we test retrieval on the MathFormula (Drechsel et al., 2025) task, and classification on MathShepherd (Wang et al., 2024b).⁵

Baselines. We compare EuroBERT with the multilingual encoders XLM-RoBERTa (Conneau et al., 2020; Goyal et al., 2021), mGTE-MLM (Zhang et al., 2024)⁶ and mDeBERTa-v3. Additionally, for code and mathematical tasks, we compare with the English-only ModernBERT (Warner et al., 2024) models.

Fine-tuning. We follow a standardized fine-tuning protocol to ensure fair model comparison. For each task, models are trained for 10,000 steps (unless otherwise specified) on the corresponding training split using a batch size of 32, a 10% warm-up ratio, and a linear learning rate decay. For small datasets requiring multiple epochs, we apply early stopping with a patience of one epoch based on validation performance. To account for model specificities, we fine-tune using 10 logarithmically spaced learning rates (1×10^{-5} to 1×10^{-4}), selecting the one that achieves the highest validation metric.⁷ For sequence classification, we use the cross-entropy loss during training, while for sequence regression, we substitute it with mean squared error.⁸ For token classification tasks, we use the token-level cross-entropy loss, assigning each sub-token in an entity to the corresponding entity label.⁹ For all retrieval tasks, we finetune for 1,000 steps on MS-MARCO (Bajaj et al., 2016)¹⁰ using the InfoNCE loss (Oord et al., 2018) with in-batch negatives and cosine similarity.

⁴<https://huggingface.co/datasets/Samoed/WikipediaRetrievalMultilingual>

⁵We detail the evaluation setup for each task in Appendix D.

⁶Since the EuroBERT models are general-purpose encoders, we evaluate them against the pre-trained mGTE-MLM variant, which, similarly, was not optimized for retrieval tasks.

⁷Additional fine-tuning hyperparameters, such as AdamW parameters (β_1 , β_2 , ϵ , and weight decay), are set according to the values reported in the original source papers. For EuroBERT models, we maintain the same settings as those used during pre-training and annealing.

⁸On summarization (SeaHorse), we train for 5,000 steps to reduce computational costs.

⁹At inference time, the final label of an entity is determined by majority vote over the sub-tokens.

¹⁰Since many retrieval datasets lack dedicated training splits, we use MS-MARCO, an English-only dataset. This choice also allows us to assess cross-lingual generalization.

Benchmark	mDeBERTa	mGTE	XLM-RoBERTa			EuroBERT		
	280M	305M	280M	560M	3.5B	210M	610M	2.1B
Retrieval ($nDCG@10$)								
MIRACL	37.5 <i>6</i> 43.7 <i>5</i>	91.2 <i>2</i> 93.8 <i>2</i>	85.4 <i>4</i> 89.5 <i>4</i>	89.4 <i>3</i> 91.6 <i>3</i>	91.4 <i>2</i> 92.6 <i>3</i>	90.8 <i>2</i> 95.1 <i>1</i>	92.6 <i>2</i> 95.0 <i>1</i>	92.9 <i>1</i> 94.8 <i>1</i>
MLDR	18.3 <i>6</i> 20.0 <i>6</i>	67.8 <i>2</i> 73.2 <i>2</i>	54.6 <i>5</i> 58.7 <i>5</i>	60.8 <i>4</i> 65.2 <i>4</i>	65.9 <i>2</i> 70.0 <i>3</i>	65.4 <i>3</i> 73.4 <i>2</i>	68.6 <i>1</i> 75.8 <i>1</i>	66.1 <i>2</i> 72.9 <i>2</i>
CC-News	18.5 <i>7</i> 15.8 <i>7</i>	71.3 <i>4</i> 71.5 <i>4</i>	61.6 <i>6</i> 60.4 <i>6</i>	72.8 <i>3</i> 72.1 <i>4</i>	80.9 <i>1</i> 80.9 <i>1</i>	67.2 <i>5</i> 69.0 <i>4</i>	75.6 <i>2</i> 76.6 <i>2</i>	75.9 <i>2</i> 76.9 <i>2</i>
Wikipedia	57.6 <i>5</i> 58.9 <i>5</i>	94.1 <i>2</i> 94.6 <i>3</i>	91.0 <i>4</i> 91.7 <i>4</i>	93.1 <i>3</i> 93.6 <i>3</i>	96.3 <i>1</i> 96.7 <i>1</i>	94.4 <i>2</i> 95.6 <i>2</i>	95.9 <i>1</i> 96.6 <i>1</i>	95.8 <i>1</i> 96.6 <i>1</i>
Sequence Classification (Accuracy)								
XNLI	79.5 <i>4</i> 82.0 <i>4</i>	75.8 <i>5</i> 78.4 <i>6</i>	74.1 <i>6</i> 76.6 <i>7</i>	81.7 <i>2</i> 84.1 <i>3</i>	83.7 <i>1</i> 86.1 <i>2</i>	76.6 <i>5</i> 79.9 <i>5</i>	81.9 <i>3</i> 84.7 <i>2</i>	84.1 <i>1</i> 86.8 <i>1</i>
PAWS-X	91.9 <i>2</i> 91.9 <i>2</i>	89.8 <i>3</i> 89.8 <i>3</i>	88.9 <i>4</i> 88.9 <i>4</i>	92.4 <i>2</i> 92.4 <i>2</i>	92.9 <i>1</i> 92.9 <i>1</i>	89.9 <i>3</i> 89.9 <i>3</i>	92.2 <i>2</i> 92.2 <i>2</i>	93.0 <i>1</i> 93.0 <i>1</i>
AmazonReviews	62.1 <i>2</i> 63.7 <i>2</i>	61.5 <i>3</i> 62.7 <i>3</i>	61.1 <i>3</i> 62.7 <i>3</i>	63.1 <i>1</i> 64.5 <i>1</i>	63.6 <i>1</i> 64.7 <i>1</i>	61.7 <i>2</i> 63.0 <i>2</i>	62.6 <i>2</i> 64.0 <i>2</i>	63.2 <i>1</i> 64.5 <i>1</i>
MassiveIntent	86.5 <i>3</i> 87.3 <i>2</i>	86.9 <i>2</i> 87.5 <i>2</i>	86.3 <i>3</i> 87.2 <i>2</i>	88.2 <i>1</i> 88.8 <i>1</i>	87.9 <i>1</i> 88.5 <i>1</i>	86.5 <i>3</i> 87.2 <i>2</i>	87.2 <i>2</i> 87.8 <i>2</i>	87.5 <i>2</i> 88.2 <i>2</i>
Token Classification (F1 Score)								
NER	96.2 <i>2</i> 96.2 <i>2</i>	95.2 <i>6</i> 95.2 <i>6</i>	95.5 <i>5</i> 95.5 <i>5</i>	96.1 <i>2</i> 96.1 <i>2</i>	96.3 <i>2</i> 96.3 <i>2</i>	94.7 <i>6</i> 94.7 <i>6</i>	95.9 <i>4</i> 95.9 <i>4</i>	95.2 <i>6</i> 95.2 <i>6</i>
Sequence Regression (Spearman)								
WMT (Ref-based)	45.7 <i>4</i> 46.5 <i>4</i>	43.9 <i>5</i> 44.0 <i>5</i>	43.0 <i>6</i> 43.1 <i>6</i>	45.3 <i>4</i> 45.6 <i>5</i>	47.7 <i>2</i> 48.5 <i>2</i>	45.2 <i>4</i> 45.1 <i>4</i>	46.0 <i>3</i> 46.5 <i>4</i>	47.3 <i>2</i> 48.5 <i>2</i>
WMT (Ref-free)	42.0 <i>3</i> 41.6 <i>2</i>	38.5 <i>5</i> 37.7 <i>5</i>	36.5 <i>7</i> 34.2 <i>6</i>	40.8 <i>4</i> 39.0 <i>4</i>	44.5 <i>1</i> 44.4 <i>1</i>	41.0 <i>3</i> 40.5 <i>3</i>	41.5 <i>3</i> 41.1 <i>3</i>	38.7 <i>5</i> 38.8 <i>4</i>
SeaHorse	64.2 <i>5</i> 60.3 <i>5</i>	63.0 <i>6</i> 59.2 <i>6</i>	61.1 <i>7</i> 56.9 <i>7</i>	65.5 <i>4</i> 61.4 <i>4</i>	67.5 <i>2</i> 63.3 <i>2</i>	63.8 <i>5</i> 60.1 <i>5</i>	66.0 <i>3</i> 62.7 <i>2</i>	67.5 <i>1</i> 64.0 <i>1</i>

Table 1: Results for multilingual tasks, with scores aggregating all languages shown above and scores aggregating European languages in *italic* below. Models are grouped into statistically significant clusters, with best ranked models highlighted in bold.

Evaluation metrics. We report accuracy for sequence classification, Spearman rank correlation for regression, F1 score for token classification, and $nDCG@10$ for retrieval tasks. We also follow Freitag et al. (2023), and group systems into language-specific clusters based on statistically significant performance gaps at 95% confidence thresholds. We then compute system-level rankings using a normalized Borda count (Colombo et al., 2022), defined as the average over the obtained per-language clusters. Note that a first cluster will only exist if a model significantly outperforms all others on a majority of languages.

3.2 Results

Table 1 presents the aggregated results across multilingual tasks, and Table 2 summarizes performance on code and mathematics benchmarks.¹¹

The EuroBERT family delivers strong performance across diverse domains and tasks. Our largest model, EuroBERT-2.1B, ranks first on 10 out of 18 tasks, competing closely with the larger XLM-RoBERTa-3.5B. EuroBERT-610M is also on par with XLM-RoBERTa-3.5B across several multilingual tasks while being five times smaller, and outperforms it on code

¹¹We provide per-language results in Appendix F.

Benchmark	ModernBERT		mDeBERTa	mGTE	XLM-RoBERTa			EuroBERT		
	150M	395M	280M	305M	280M	560M	3.5B	210M	610M	2.1B
Code Retrieval ($nDCG@10$)										
CodeSearchNet	53.9 5	65.8 3	2.8	34.0 7	23.0 8	40.8 6	54.1 5	58.9 4	69.9 2	72.6 1
DupStackMath	39.7 4	45.5 2	10.2 7	37.5 4	29.3 6	36.9 5	42.9 3	41.7 3	46.0 2	48.3 1
Code Classification (Accuracy)										
CodeComplexity	86.1 3	88.6 3	73.9 5	74.5 5	74.1 5	83.6 4	84.3 4	91.9 2	94.2 1	95.2 1
CodeDefect	65.8 3	67.0 2	64.7 3	63.5 4	61.9 4	54.3 5	65.8 3	69.5 1	69.0 1	67.7 2
Math Retrieval ($nDCG@10$)										
MathFormula	89.6 5	91.9 2	85.2 7	83.4 8	83.1 8	81.4	89.1 6	91.5 3	92.6 1	91.0 4
Math Classification (Accuracy)										
MathShepherd	77.7 4	83.6 2	75.1 5	77.2 4	71.9 6	67.6 7	82.5 3	84.0 2	87.3 1	86.8 1

Table 2: Results for code and mathematical tasks. Models are grouped into statistically significant clusters, with best ranked models highlighted in bold.

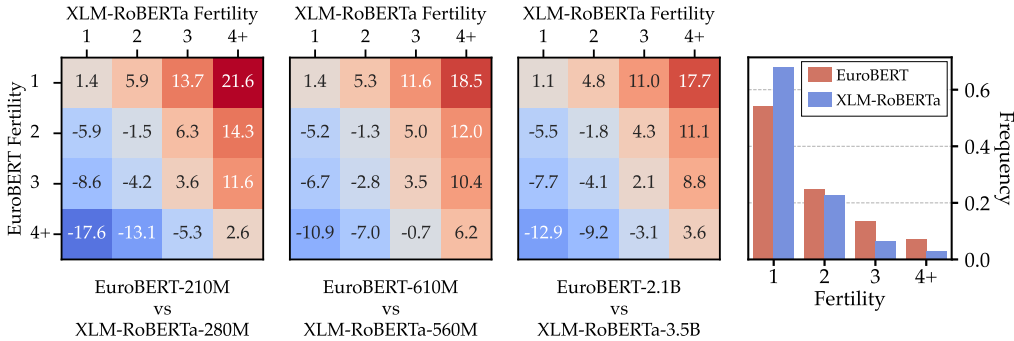


Figure 2: Difference in F1 Score between EuroBERT and XLM-RoBERTa by tokenizer fertility (left plot) and fertility distribution for both tokenizers (right plot) on the NER dataset.

and mathematics benchmarks. Likewise, EuroBERT-210M matches XLM-RoBERTa-560M performance while having less than half the parameters, and consistently outperforms other models of similar size, showing especially strong results on European languages.

EuroBERT is effective at document ranking. Across domains, EuroBERT consistently ranks high for retrieval tasks. Notably, the 210M and 610M parameter models outperform all models of comparable sizes, and are competitive with the larger XLM-RoBERTa-3.5B.¹²

EuroBERT models are on par with similarly sized models for sequence classification. On sequence classification, no model significantly outperforms all others. During the development of EuroBERT, we found that several design decisions lead to a trade-off between retrieval and classification capabilities (§4). We highlight, however, that EuroBERT-2.1B is still among the highest ranking systems, and that the smaller models in the family are competitive with models of comparable size.

There is room for improvement on token classification. On the NER task, EuroBERT lags behind XLM-RoBERTa. However, we observed that models achieve comparable F1 scores when splitting entities into a similar number of tokens, as shown in Figure 2. In contrast, models are significantly better when dividing entities into fewer tokens. We posit that token

¹²For retrieval, increasing model size did not always lead to better results. Further analysis, in Appendix E, revealed that EuroBERT-2.1B benefits significantly from a more thorough grid search.

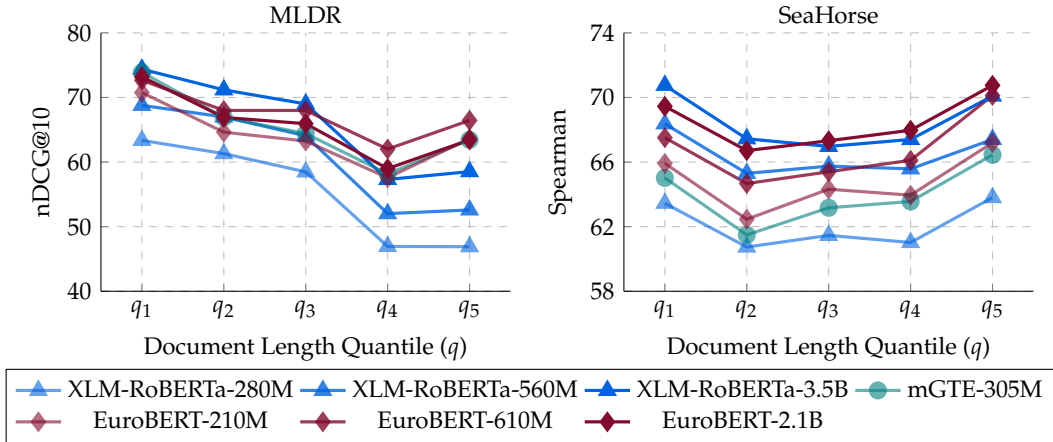


Figure 3: Results by length of the positive documents for retrieval (MLDR) and input documents for summarization (SeaHorse).

classification tasks may benefit from larger vocabularies, with lower fertility, such as the one used in the XLM-RoBERTa family. We also note, however, that increasing the vocabulary size also increases the number of parameters, particularly for smaller models.¹³

EuroBERT can function as an evaluation metric. EuroBERT models match or exceed the performance of similarly sized systems in reference-based translation evaluation. For reference-free evaluation, while EuroBERT-2.1B lags behind the larger XLM-RoBERTa, the 210M and 610M variants are competitive with other baselines. In the future, we will explore other training signals to further enhance EuroBERT’s cross-lingual capabilities. For summary evaluation, EuroBERT models consistently outperform similarly sized alternatives.

EuroBERT maintains performance at longer context lengths. Figure 3 compares the long context performance of EuroBERT and XLM-RoBERTa. While both models achieve similar performance for shorter inputs, EuroBERT maintains performance at longer contexts, whereas XLM-RoBERTa suffers notable degradation.

The EuroBERT family excels in tasks related to code and mathematics. On these tasks in the code and math domain, all EuroBERT models consistently surpass other systems. Notably, EuroBERT-210M reflects most of the performance of the larger models in the family, and ranks above all baselines, highlighting its capabilities at a smaller scale.

4 Training Recipe Analysis

We measure the impact of various design decisions made during the development of EuroBERT with extensive ablations. Following Blakeney et al. (2024) and Llama Team (2024), we perform multiple annealing runs on 40B tokens, each varying a different component of our recipe, and measure the performance on the XNLI and MIRACL validation sets, the former representing multilingual classification and the latter multilingual retrieval.¹⁴

Balancing the language distribution enhances performance. The left-most plot in Figure 4 reports retrieval and classification performance as the proportion of English is reduced

¹³For example, doubling the vocabulary size of EuroBERT-210M would add 100M parameters to the model embeddings.

¹⁴We follow the procedure from §3, but instead evaluate on the validation splits considering only European languages.

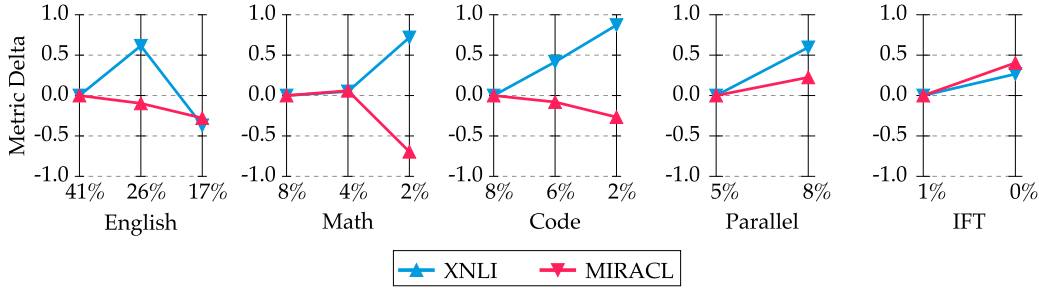


Figure 4: Impact of changing data subset ratios during annealing. The first vertical axis in each subplot denotes the *reference* data mix from Table 6.

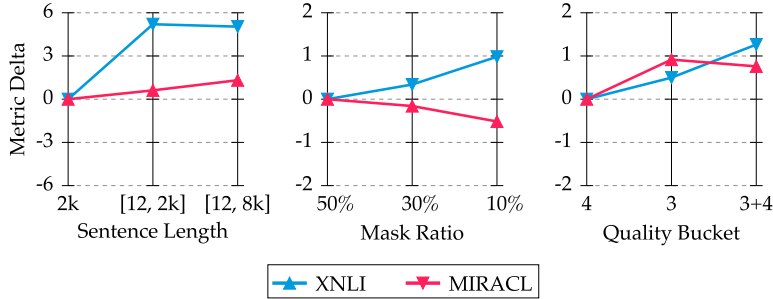


Figure 5: Impact of hyperparameter choices during annealing. The first vertical axis in each subplot denotes the *reference* data mix from Table 6.

and re-distributed between other languages. We observe that a more balanced distribution improves overall results. However, when the language distribution becomes too close to uniform, there is a degradation in performance.

Including math and code improves multilingual retrieval, but degrades multilingual classification. The second and third plots in Figure 4 show MIRACL performance dropping and XNLI accuracy rising as the quantity of math and code data decreases. In future work, we will investigate how to better balance downstream task performance during pre-training.

Increasing parallel data improves multilingual classification and retrieval. The forth plot in Figure 4 presents XNLI and MIRACL performance when increasing parallel data. In line with recent work showing the benefits of pre-training with parallel data (Anil et al., 2023; Briakou et al., 2023; Alves et al., 2024), we find it improves both benchmarks.

Adding instruction fine-tuning data degrades model performance. The right-most plot in Figure 4 analyses the impact of adding instructions during annealing, which can improve performance for decoder language models (Wei et al., 2022; Chung et al., 2024). In contrast to decoders, it leads to worse performance when training an encoder model.

Varying sentence length improves performance. The first plot in Figure 5 examines the impact of variable sentence lengths during annealing. Compared to the fixed packed sentence lengths employed in pretraining, variable sentence lengths significantly boosts XNLI and moderately MIRACL performance.¹⁵ This improvement remains stable, without degradation when the maximum context length is extended to 8,192 tokens.

A reduced masking ratio during annealing enhances classification accuracy. Similarly to previous research advocating a lower masking ratio in later training (Yang et al., 2023;

¹⁵We hypothesize this gap stems from the prevalence of shorter sequences in the XNLI dataset.

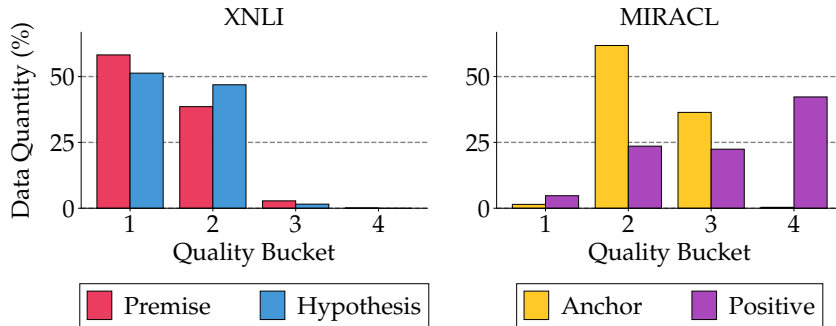


Figure 6: Quality buckets for XNLI and MIRACL English training subset.

Ankner et al., 2024), we also find that reducing it to 10% during the annealing phase improves EuroBERT’s accuracy on XNLI, though it leads to a decline in MIRACL scores.

Filtering data based on educational value can degrade results. Contrary to initial expectations, using the highest quality data bucket during annealing does not result in better performance on XNLI and MIRACL. Instead, as illustrated in the right-most plot of Figure 5, mixing the buckets with quality levels 3 and 4 leads to the best overall results. Curiously, inspecting the evaluation splits of these datasets revealed that our quality filter would discard nearly all examples in XNLI, and many in MIRACL, as shown in Figure 6. This result highlights a potential domain mismatch, wherein our training data deviates from the distribution of downstream tasks. Indeed, while educational value fits assistant-like tasks typically delegated to LLMs, a broader coverage of textual data, reflected in mixing both quality buckets, may be more inline with general-purpose vector representations. In future work, we would like to explore quality filters that are better tailored to encoders.

Final annealing configuration. The previous results revealed several design choices that trade off classification and retrieval performance. In the final data mix, we aimed to balance these two tasks. Based on the previous analysis, we created our final annealing dataset by selecting data above the third threshold. We reduced the proportion of English to 26% while proportionally increasing the share of the remaining languages. We allocated 6% and 4% of the data mix to math and code, respectively. Additionally, we increased the proportion of parallel data to 6%, and removed instructions. We finally lowered the masking ratio to 10% and performed annealing with random sentence lengths of up to 8,192 tokens.

5 Related Work

Encoder models have shown strong performance in non-generative tasks, such as classification and retrieval (Devlin et al., 2019; Liu et al., 2019; He et al., 2023; Acheampong et al., 2021; Ma et al., 2019; Karpukhin et al., 2020; Wang et al., 2024a). Variants of these models have also extended support to multiple languages and cross-lingual tasks (Conneau et al., 2020). However, scaling to many languages introduces the “curse of multilinguality” (Conneau et al., 2020; Chang et al., 2024), where interference across languages degrades performance. Notably, increasing model capacity has been shown to mitigate this effect (Conneau et al., 2020), motivating our focus on scale.

Encoders are typically trained with masked language modeling (MLM) (Devlin et al., 2019). While alternatives like replaced token detection (He et al., 2023) exist, we adopt the MLM objective because initial evaluations of existing models showed more balanced results across tasks. Building upon the effectiveness of higher masking ratios (Wettig et al., 2023), we mask 50% of the training tokens during pre-training. Prior work has also shown the benefits of decreasing the masking ratio in later phases of training (Yang et al., 2023; Ankner et al., 2024), we also lower our masking ratio to 10% during annealing. Interestingly, we demonstrate that this choice improves classification accuracy, but reduces retrieval quality.

Recent concurrent work, such as ModernBERT (Warner et al., 2024) and mGTE (Zhang et al., 2024), also revisits encoders in light of advances in decoder models. Similar to our approach, they incorporate grouped query attention (Ainslie et al., 2023), rotary positional embeddings (Su et al., 2024), gated linear units (Shazeer, 2020), root mean square layer normalization (Zhang & Sennrich, 2019), and support for longer context windows. However, we additionally draw inspiration from Llama Team (2024); Yang et al. (2024) by including code and mathematical data during pre-training, which we show improves retrieval quality.

6 Conclusion

We propose a recipe for training general-purpose multilingual encoders, creating the EuroBERT family. We incorporate recent architectural advances from decoder models, and train on a multilingual dataset containing European and globally spoken languages, together with code and mathematics. Our models outperform existing alternatives on a comprehensive suite of tasks covering multilingual capabilities, mathematics and code. We also extensively analyze the design decisions behind EuroBERT’s dataset and training pipeline. Alongside this paper, we release all models in the EuroBERT family, including intermediate training checkpoints, and our training framework to facilitate future research.

Acknowledgments

We sincerely thank the ADAstra supercomputer (CINES) for its technical support and high-performance computing (HPC) resources, provided through grants C1615122 and GDA2401. We also appreciate the support of the French government through the France 2030 program as part of the ArGiMi project. This work was also supported by the EU’s Horizon Europe Research and Innovation Actions (UTTER, contract 101070631), by the project DECOLLAGE (ERC-2022-CoG 101088763), by the Portuguese Recovery and Resilience Plan through project C645008882-00000055 (Center for Responsible AI), and by FCT/MECI through national funds and when applicable co-funded EU funds under UID/50008: Instituto de Telecomunicações. Duarte was also partially supported by the DataIA Institute, whose contributions facilitated the completion of this work.

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A EuroBERT Model Architecture

Table 3 reports the architectural details of the EuroBERT model family.

Model Size	210M	610M	2.1B
Layers	12	26	32
Embedding Dimension	768	1,152	2,304
FFN Dimension	3,072	4,096	6,144
Attention Heads	12	18	18
Key/Value Heads	12	6	6
Layer Normalization	RMSNorm		
RMSNorm ϵ	1×10^{-5}		
Activation Function	SwiGLU		
Vocabulary Size	128,000		
Positional Embeddings	RoPE		
RoPE θ	250,000		
Tokenizer	LLaMA 3		

Table 3: Summary of architectural hyperparameters for EuroBERT models of different sizes.

B Training Details

We trained the EuroBERT family on Adastral, utilizing 92 MI250X GPUs for EuroBERT-210M over 15k hours, 384 MI250X GPUs for EuroBERT-610M over 92k hours, and 96 MI300A GPUs for EuroBERT-2.1B over 106k hours, hyperparameter selections are detailed in Table 4. We find this training recipe highly stable, with no loss spikes or need for intervention to address model training divergence (Figure 7).

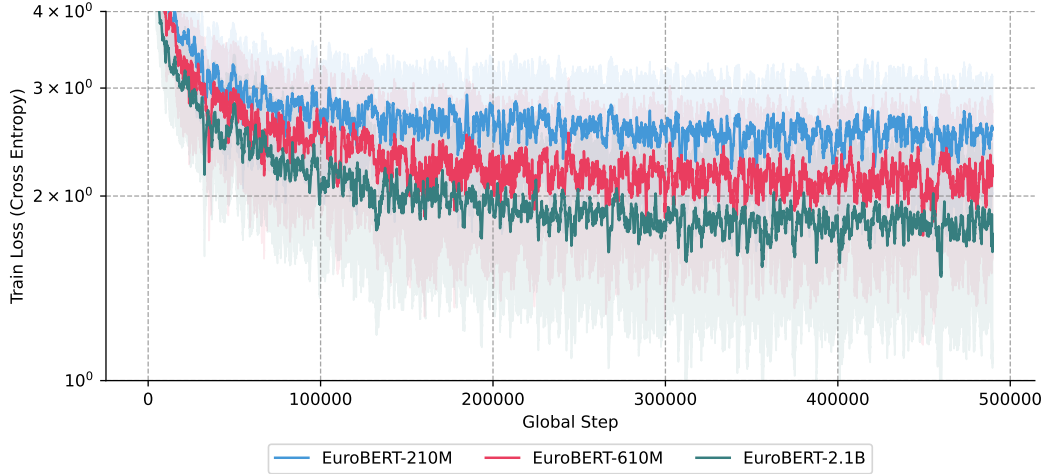


Figure 7: Pre-training Loss for all EuroBERT models on a logarithmic scale.

Parameter	210M	610M	2.1B
Pre-training			
LR		1e-4	
LR Scheduler		WSD	
Warmup Steps		2,000	
Context Length		2,048	
Weight Initialisation	$\mathcal{N}(\mu = 0, \sigma^2 = 0.2)$		
Annealing			
LR		1e-4 to 0	
LR Scheduler		Cosine	
Context Length		8,192	
Optimizer			
Optimizer		AdamW	
Beta1		0.9	
Beta2		0.95	
Epsilon (eps)		1e-5	
Weight Decay		0.1	
Clip Grad Norm		1.0	
Training Setup			
Per-GPU Batch Size	24	12	10
Gradient Accumulation Steps	1	1	5
GPUs	192	384	96
Tokens/Step	9,437,184	9,437,184	9,830,400

Table 4: Training hyperparameters for EuroBERT models (210M, 610M, 2.1B). The optimizer and Tokens/Step remain consistent across both pre-training and annealing phases.

C Data Mix

Table 5 details the data sources used throughout training, as well as the number of tokens used from each of the data source during pre-training. Table 6 specifies the different annealing mixes used for ablations.

Source	Subset	Tokens (M)	Mix (%)	Source	Subset	Tokens (M)	Mix (%)
FineWeb	English	2,002,327	41.34	The-Stack v2	JavaScript	58,440	1.21
CulturaX	French	295,113	6.09	The-Stack v2	PHP	25,620	0.53
CulturaX	German	291,514	6.02	The-Stack v2	C#	24,842	0.51
CulturaX	Spanish	290,489	6.00	The-Stack v2	Python	21,521	0.44
CulturaX	Chinese	238,467	4.92	The-Stack v2	Java	20,950	0.43
CulturaX	Italian	120,128	2.48	The-Stack v2	Go	14,766	0.30
CulturaX	Russian	116,797	2.41	The-Stack v2	TypeScript	11,307	0.23
CulturaX	Portuguese	112,321	2.32	The-Stack v2	HTML	7,962	0.16
CulturaX	Japanese	112,242	2.32	The-Stack v2	Lua	7,733	0.16
CulturaX	Polish	111,659	2.31	The-Stack v2	Ruby	5,524	0.11
CulturaX	Turkish	53,126	1.10	The-Stack v2	Vue	5,411	0.11
CulturaX	Arabic	52,413	1.08	The-Stack v2	R	5,287	0.11
CulturaX	Vietnamese	50,661	1.05	The-Stack v2	Shell	4,793	0.10
CulturaX	Dutch	50,646	1.05	The-Stack v2	Swift	3,766	0.08
CulturaX	Hindi	25,544	0.53	The-Stack v2	reStructuredText	3,761	0.08
EuroLLM Parallel	es ↔ en	50,613	1.05	The-Stack v2	JSON	3,586	0.07
EuroLLM Parallel	fr ↔ en	44,891	0.93	The-Stack v2	Rust	3,152	0.07
EuroLLM Parallel	de ↔ en	30,541	0.63	The-Stack v2	YAML	2,716	0.06
EuroLLM Parallel	it ↔ en	18,702	0.39	The-Stack v2	Dart	2,678	0.06
EuroLLM Parallel	ru ↔ en	13,808	0.29	The-Stack v2	RMarkdown	2,058	0.04
EuroLLM Parallel	nl ↔ en	12,666	0.26	The-Stack v2	HCL	1,423	0.03
EuroLLM Parallel	pl ↔ en	7,280	0.15	The-Stack v2	PowerShell	1,027	0.02
EuroLLM Parallel	ar ↔ en	6,414	0.13	The-Stack v2	VBA	1,027	0.02
EuroLLM Parallel	zh ↔ en	6,206	0.13	The-Stack v2	AsciiDoc	970	0.02
EuroLLM Parallel	cs ↔ en	5,458	0.11	The-Stack v2	Groovy	540	0.01
EuroLLM Parallel	hu ↔ en	4,599	0.09	The-Stack v2	CUDA	406	0.01
EuroLLM Parallel	vi ↔ en	3,395	0.07	The-Stack v2	Dockerfile	281	0.01
EuroLLM Parallel	tr ↔ en	2,975	0.06	The-Stack v2	Cython	103	0.01
EuroLLM Parallel	ja ↔ en	2,687	0.06	The-Stack v2	COBOL	96	0.01
EuroLLM Parallel	hi ↔ en	1,136	0.02	The-Stack v2	GraphQL	83	0.01
Proof-pile-2	Arxiv	121,503	2.51	The-Stack v2	HTTP	82	0.01
Proof-pile-2	Open-Web-Math	54,168	1.12	The-Stack v2	ABAP	71	0.01
Proof-pile-2	Algebraic-stack	35,985	0.74	The-Stack v2	RDoc	16	0.01
The-Stack v2	C++	120,085	2.48	The-Stack v2	Metal	8	0.01
The-Stack v2	SQL	75,348	1.56	The-Stack v2	AppleScript	7	0.01
The-Stack v2	C	59,404	1.23	Total		4,843,357	100

Table 5: Pre-training data, with a total of 4.8 trillion tokens (as measured by EuroBERT’s tokenizer). We report the list of all dataset names and subsets used, including the number of tokens selected and their proportion in the final data mix.

Mix	en	fr	de	nl	hi	it	ja	pl	pt	ru	es	ar	zh	tr	Code	Math	Parallel	IFT
Reference	46.3	5.8	5.7	1.0	0.3	1.5	0.8	1.0	1.4	1.0	5.7	0.4	4.7	1.0	8.7	8.2	5.2	1.2
English 26%	26.0	6.0	6.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	6.0	4.0	6.0	4.0	4.0	4.0	5.0	1.0
English 17%	17.0	6.0	6.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	6.0	5.0	6.0	5.0	4.0	4.0	5.0	1.0
Math 4%	46.3	5.8	5.7	1.0	0.3	1.5	0.8	1.0	1.4	1.0	5.7	0.4	4.7	1.0	8.7	4.0	5.2	1.2
Math 2%	46.3	5.8	5.7	1.0	0.3	1.5	0.8	1.0	1.4	1.0	5.7	0.4	4.7	1.0	8.7	2.0	5.2	1.2
Code 8%	46.3	5.8	5.7	1.0	0.3	1.5	0.8	1.0	1.4	1.0	5.7	0.4	4.7	1.0	6.0	8.2	5.2	1.2
Code 4%	46.3	5.8	5.7	1.0	0.3	1.5	0.8	1.0	1.4	1.0	5.7	0.4	4.7	1.0	4.0	8.2	5.2	1.2
Code 2%	46.3	5.8	5.7	1.0	0.3	1.5	0.8	1.0	1.4	1.0	5.7	0.4	4.7	1.0	2.0	8.2	5.2	1.2
Parallel 8%	46.3	5.8	5.7	1.0	0.3	1.5	0.8	1.0	1.4	1.0	5.7	0.4	4.7	1.0	8.7	8.2	8.0	1.2
IFT 0%	46.3	5.8	5.7	1.0	0.3	1.5	0.8	1.0	1.4	1.0	5.7	0.4	4.7	1.0	8.7	8.2	5.2	0.0

Table 6: Data mix employed in the ablation study measuring the importance of different data subsets in the EuroBERT annealing phase.

D Details of Evaluation Datasets

This appendix offers additional details on the datasets used for evaluation. Table 7 presents the language coverage of all evaluation datasets, and below are additional specifications on the evaluated tasks.

Task	European Languages								Extra-European Languages								Code	Math
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh			
Information Retrieval																		
MIRACL	✓		✓	✓						✓	✓	✓	✓			✓		
MLDR	✓	✓	✓	✓	✓			✓		✓	✓	✓				✓		
Wikipedia	✓	✓			✓	✓		✓				✓						
CC-News	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
CodeSearchNet	✓															✓		
DupStackMath	✓															✓		
MathFormula	✓															✓		
Sequence Classification																		
XNLI	✓	✓	✓	✓					✓	✓		✓	✓	✓	✓			
PAWS-X	✓	✓	✓	✓														
AmazonReviews	✓	✓	✓	✓							✓					✓		
MassiveIntent	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
CodeDefect	✓															✓		
CodeComplexity	✓															✓		
MathShepherd	✓															✓		
Sequence Regression																		
WMT	✓	✓		✓			✓				✓	✓	✓		✓			
SeaHorse	✓	✓	✓										✓	✓				
Token Classification																		
NER	✓	✓	✓			✓												

Table 7: Language coverage across evaluation datasets.

Retrieval datasets:

- **MS-MARCO** (Bajaj et al., 2016) — English-only retrieval dataset used for fine-tuning, where each anchor-positive pair includes a mined hard negative, forming a triplet structure.
- **MIRACL** (Zhang et al., 2023) — Multilingual retrieval dataset. We use the semi-supervised version with labeled positive pairs provided by SentenceTransformers¹⁶ as the primary data source. Anchors serve as queries, and the corpus consists of all positive documents in the dataset. Since only a single data split is available, we create validation and test sets by partitioning 50% of the original split for each, using queries as the split key to ensure no data leakage.
- **MLDR** (Chen et al., 2024) — Long-context multilingual retrieval dataset. As with MIRACL, we use the triplet version provided by SentenceTransformers and apply the same validation-test split strategy.
- **Wikipedia**¹⁷ — Multilingual information retrieval dataset. Since only a single data split is available, we partition 50% of the queries into validation and test sets.
- **CC-News** (de Gibert et al., 2024) — Highly multilingual retrieval dataset. As with MIRACL, we use the SentenceTransformers dataset version as the primary data source and apply the same test-validation split method.
- **CodeSearchNet** (Husain et al., 2019) — Code retrieval dataset with comment-code query-positive pairs (SentenceTransformers version), processed similarly to the previous datasets.

¹⁶<https://huggingface.co/collections/sentence-transformers/embedding-model-datasets-6644d7a3673a511914aa7552>

¹⁷<https://huggingface.co/datasets/Samoed/WikipediaRetrievalMultilingual>

- **DupStackMath** (Hoogeveen et al., 2015) — Code retrieval dataset with queries, a corpus, and relevant documents, processed the same way as the above datasets.
- **MathFormula** (Drechsel et al., 2025) — Mathematical retrieval dataset consisting of pairs of equivalent formulas. The original dataset contains formula pairs labeled as true or false based on their equivalence, spanning 71 well-known mathematical formulas. To construct the retrieval dataset, we extract only equivalent formula pairs, retaining positive instances. Due to the dataset’s large size, we sample 100 positive pairs per formula type for both validation and test sets. The final dataset is processed following the same methodology as other pair-based datasets.

Sequence classification datasets:

- **XNLI** (Conneau et al., 2018) — Natural language inference task extending MNLI (Williams et al., 2018) to non-English languages, consisting in classifying sentence pairs into entailment, contradiction, or neutral.
- **PAWS-X** (Yang et al., 2019) — Paraphrase identification task aimed at determining whether two sentences convey the same meaning. Fine-tuning is performed cross-lingually, with training on the English subset and evaluation across all available languages.
- **AmazonReviews** (Keung et al., 2020) — Sentiment analysis task consisting in estimating the satisfaction level of multilingual Amazon product reviews on a 1-to-5 scale. Fine-tuning is performed on all available languages.
- **MassiveIntent** (Keung et al., 2020) — Multilingual classification task consisting in assigning sentences to one of 60 topic categories. Fine-tuning is performed on all available languages.
- **CodeDefect** (Zhou et al., 2019) — Binary classification task aimed at identifying whether a given code snippet contains a defect.
- **CodeComplexity** (Jeon et al., 2023) — Computational analysis task consisting in estimating the order of complexity of a code-formulated computer science problem.
- **MathShepherd** (Wang et al., 2024b) — Binary classification task aimed at determining whether a step-by-step math rationale is correct given a problem prompt. We limited the dataset to rationales with 3 steps to mitigate the class imbalance observed in longer rationales, where incorrect solutions become more frequent. As the dataset lacks a validation split, we allocate half of the test set for validation.

Sequence regression datasets:

- **WMT** (Bojar et al., 2017; 2018; Barrault et al., 2019; 2020; Akhbardeh et al., 2021; Kocmi et al., 2022) — Regression task consisting in estimating translation quality given a source sentence, and possibly a reference translation. As the original test set covers only three language pairs, we construct validation and test sets by sampling 5% of the training set for each, ensuring broader language coverage in evaluation. We report results under both the reference-free and reference-based evaluation settings.
- **SeaHorse** (Clark et al., 2023) — Multilingual summarization evaluation task, where each text-summary pair is annotated across 6 binary evaluation dimensions. The final score is obtained by averaging these labels, yielding a continuous value between 0 and 1. To avoid penalizing models with limited context lengths, the summary is placed first in the input, followed by the main text, ensuring the model can attend to the full summary.

Token classification datasets:

- **NER** (Liang et al., 2020) — Named entity recognition task from the XGLUE benchmark, combining subsets of CoNLL2002 (Tjong Kim Sang, 2002) and CoNLL2003 (Tjong Kim Sang & De Meulder, 2003), adapted for cross-lingual evaluation with English-only training. It spans four languages (English, German, Spanish, and Dutch) and targets four entity types (Person, Location, Organization, and Miscellaneous).

E Results on larger hyper-parameter search

In this appendix, we investigate why the largest EuroBERT model does not significantly outperform its 610M counterpart on retrieval tasks. We hypothesize the larger model may benefit from a more comprehensive hyper-parameter search. To test this hypothesis, we perform a more extensive grid search for fine-tuning on the MS-MARCO dataset across the EuroBERT family. Rather than varying only the learning rate with fixed optimization settings, we systematically explore a broader set of hyperparameters: Adam’s β_2 (0.95 [default], 0.98, 0.999), Adam’s ϵ (10^{-5} [default], 10^{-8}), and the number of training steps (1,000 [default], 2,000).

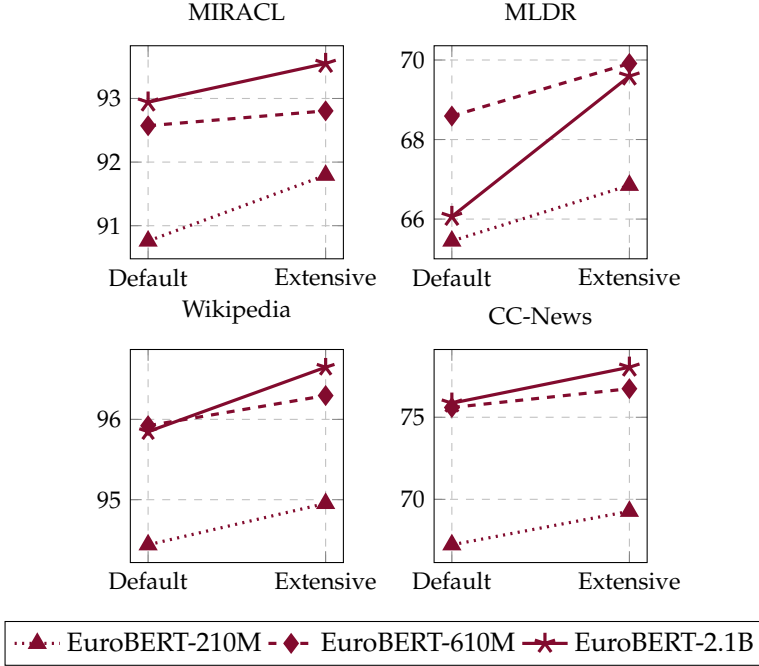


Figure 8: **Retrieval performance** of EuroBERT models under the default fine-tuning configuration (Default) compared to the more extensive hyperparameter grid search (Extensive). Results are reported as average nDCG@10 across supported languages.

Figure 8 demonstrates that, while all models benefit from a denser hyperparameter grid search, the largest EuroBERT model exhibits the most substantial improvements, particularly on the MIRACL and MLDR datasets. Additionally, as shown in Table 8, increasing the number of training steps from 1,000 to 2,000 consistently enhances performance across all model sizes. Also, we observe that models generally benefit from increasing β_2 and reducing the ϵ and learning rate.

MIRACL								
Model	Learning Rate		Adam β_2		Adam ϵ		Steps	
	Default	Extensive	Default	Extensive	Default	Extensive	Default	Extensive
EuroBERT-210M	4.6e-05	2.8e-05	0.95	0.98	1e-05	1e-05	1,000	2,000
EuroBERT-610M	3.6e-05	2.8e-05	0.95	0.98	1e-05	1e-08	1,000	1,000
EuroBERT-2.1B	3.6e-05	1.7e-05	0.95	0.98	1e-05	1e-08	1,000	2,000

MLDR								
Model	Learning Rate		Adam β_2		Adam ϵ		Steps	
	Default	Extensive	Default	Extensive	Default	Extensive	Default	Extensive
EuroBERT-210M	2.8e-05	2.8e-05	0.95	0.98	1e-05	1e-05	1,000	2,000
EuroBERT-610M	2.2e-05	2.8e-05	0.95	0.95	1e-05	1e-05	1,000	2,000
EuroBERT-2.1B	4.6e-05	1.3e-05	0.95	0.98	1e-05	1e-08	1,000	2,000

Wikipedia								
Model	Learning Rate		Adam β_2		Adam ϵ		Steps	
	Default	Extensive	Default	Extensive	Default	Extensive	Default	Extensive
EuroBERT-210M	2.8e-05	2.2e-05	0.95	0.98	1e-05	1e-08	1,000	2,000
EuroBERT-610M	3.6e-05	2.2e-05	0.95	0.95	1e-05	1e-08	1,000	2,000
EuroBERT-2.1B	2.8e-05	2.8e-05	0.95	0.95	1e-05	1e-05	1,000	2,000

CC-News								
Model	Learning Rate		Adam β_2		Adam ϵ		Steps	
	Default	Extensive	Default	Extensive	Default	Extensive	Default	Extensive
EuroBERT-210M	4.6e-05	3.6e-05	0.95	0.98	1e-05	1e-05	1,000	2,000
EuroBERT-610M	4.6e-05	2.8e-05	0.95	0.95	1e-05	1e-05	1,000	2,000
EuroBERT-2.1B	3.6e-05	3.6e-05	0.95	0.95	1e-05	1e-05	1,000	2,000

Table 8: Overview of the optimal hyperparameters selected on the validation set for both the Default and Extensive fine-tuning configurations.

F Detailed Results

Table 9 and Table 10 present per-language results for the retrieval and sequence classification tasks, respectively. Table 11 and Table 12 report detailed performance on sequence regression. Finally, Table 13 provides per-language results on token classification (NER task).

MIRACL																	
Model	European Languages								Extra-European Languages								Average
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World
XLM-RoBERTa-280M	88.0	—	88.6	91.9	—	—	—	—	84.6	77.9	85.7	82.5	—	—	83.8	89.5	85.4
XLM-RoBERTa-560M	89.7	—	91.6	93.5	—	—	—	—	89.0	81.1	91.1	88.6	—	—	90.3	91.6	89.4
XLM-RoBERTa-3.5B	90.4	—	93.0	94.5	—	—	—	—	91.5	85.1	92.6	92.1	—	—	91.9	92.6	91.4
mDeBERTa-v3-280M	45.3	—	39.6	46.2	—	—	—	—	34.4	36.0	33.7	29.6	—	—	35.3	43.7	37.5
mGTE-MLM-305M	91.4	—	94.6	95.2	—	—	—	—	91.6	85.3	91.5	88.3	—	—	91.4	93.8	91.2
ModernBERT-150M	93.4	—	71.2	81.2	—	—	—	—	8.6	3.0	36.0	21.3	—	—	45.6	81.9	45.0
ModernBERT-395M	95.1	—	84.2	90.7	—	—	—	—	5.7	8.7	32.7	19.2	—	—	34.3	90.0	46.3
EuroBERT-210M	94.1	—	95.4	95.8	—	—	—	—	90.0	83.2	90.8	85.7	—	—	90.9	95.1	90.8
EuroBERT-610M	93.6	—	95.1	96.3	—	—	—	—	91.8	88.3	92.4	90.7	—	—	92.4	95.0	92.6
EuroBERT-2.1B	94.2	—	95.0	95.3	—	—	—	—	93.0	87.1	93.4	91.5	—	—	94.1	94.8	92.9

MLDR																	
Model	European Languages								Extra-European Languages								Average
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World
XLM-RoBERTa-280M	59.4	56.7	58.0	64.5	53.4	—	—	60.3	44.4	56.4	50.3	43.6	—	—	53.8	58.7	54.6
XLM-RoBERTa-560M	63.4	61.1	66.9	71.1	61.9	—	—	67.1	51.5	60.3	54.9	51.5	—	—	58.9	65.2	60.8
XLM-RoBERTa-3.5B	68.9	66.1	72.7	73.5	67.5	—	—	71.0	56.5	61.8	62.6	60.8	—	—	63.5	70.0	65.9
mDeBERTa-v3-280M	18.8	24.3	15.4	23.9	18.2	—	—	19.0	12.3	20.5	17.4	13.2	—	—	18.1	20.0	18.3
mGTE-MLM-305M	63.5	68.7	79.5	78.2	71.4	—	—	78.1	55.7	66.2	62.4	60.8	—	—	60.9	73.2	67.8
ModernBERT-150M	61.0	11.3	23.5	25.8	18.4	—	—	19.7	0.7	0.7	2.9	2.5	—	—	0.8	26.6	15.2
ModernBERT-395M	68.4	25.3	42.6	58.0	21.2	—	—	37.1	0.5	2.0	3.5	3.7	—	—	0.9	42.1	23.9
EuroBERT-210M	67.2	68.1	78.2	80.0	68.9	—	—	77.9	52.1	51.3	60.8	59.1	—	—	56.4	73.4	65.4
EuroBERT-610M	72.5	69.5	80.3	79.8	73.9	—	—	79.0	55.5	60.9	61.6	62.5	—	—	59.0	75.8	68.6
EuroBERT-2.1B	72.5	65.4	77.6	77.6	69.2	—	—	75.0	53.0	58.1	61.5	59.3	—	—	57.6	72.9	66.1

Wikipedia																	
Model	European Languages								Extra-European Languages								Average
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World
XLM-RoBERTa-280M	94.7	91.2	—	—	91.5	90.2	—	90.8	—	87.4	—	—	—	—	—	91.7	91.0
XLM-RoBERTa-560M	95.4	93.5	—	—	93.4	93.4	—	92.5	—	90.7	—	—	—	—	—	93.6	93.1
XLM-RoBERTa-3.5B	97.9	96.5	—	—	96.6	96.3	—	96.0	—	94.5	—	—	—	—	—	96.7	96.3
mDeBERTa-v3-280M	66.1	60.4	—	—	53.6	57.6	—	56.5	—	51.3	—	—	—	—	—	58.9	57.6
mGTE-MLM-305M	96.7	93.9	—	—	94.7	93.9	—	93.6	—	92.0	—	—	—	—	—	94.6	94.1
ModernBERT-150M	97.3	56.5	—	—	60.5	57.2	—	67.0	—	5.7	—	—	—	—	—	67.7	57.4
ModernBERT-395M	98.2	69.9	—	—	70.0	67.4	—	83.2	—	12.0	—	—	—	—	—	77.7	66.8
EuroBERT-210M	97.7	94.7	—	—	94.5	95.6	—	95.7	—	88.6	—	—	—	—	—	95.6	94.4
EuroBERT-610M	98.3	95.9	—	—	96.0	96.6	—	96.1	—	92.6	—	—	—	—	—	96.6	95.9
EuroBERT-2.1B	99.0	96.1	—	—	96.0	96.0	—	95.9	—	92.0	—	—	—	—	—	96.6	95.8

CC-News																	
Model	European Languages								Extra-European Languages								Average
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World
XLM-RoBERTa-280M	69.9	57.8	55.8	57.5	57.2	66.8	55.1	63.1	72.2	31.1	75.7	76.2	47.8	61.5	77.1	60.4	61.6
XLM-RoBERTa-560M	77.3	68.6	69.1	70.1	70.8	75.9	69.6	75.3	82.8	51.0	82.3	83.3	61.1	73.7	81.2	72.1	72.8
XLM-RoBERTa-3.5B	84.4	77.4	79.7	79.1	79.6	83.8	79.5	84.0	88.1	59.5	87.4	88.8	72.1	82.9	86.8	80.9	80.9
mDeBERTa-v3-280M	25.0	15.7	11.9	12.4	13.1	20.4	12.4	15.8	23.1	4.7	33.6	23.6	10.7	20.4	34.8	15.8	18.5
mGTE-MLM-305M	76.1	68.7	72.8	70.1	68.4	76.5	65.1	74.3	79.6	32.5	85.1	83.3	56.7	72.3	88.2	71.5	71.3
ModernBERT-150M	75.6	16.1	15.6	14.4	15.0	29.6	7.0	10.9	2.3	1.8	6.7	2.5	8.5	5.2	10.0	23.0	14.7
ModernBERT-395M	84.9	21.5	33.4	41.3	20.0	36.2	4.2	27.8	2.2	2.0	9.5	4.1	9.6	10.0	9.9	33.6	21.1
EuroBERT-210M	80.0	66.9	69.2	69.7	65.9	73.5	57.8	69.1	76.7	17.0	82.4	79.7	52.1	57.4	90.9	69.0	67.2
EuroBERT-610M	84.0	72.9	76.4	75.5	73.8	79.6	70.9	79.9	84.0	49.7	84.9	85.1	62.4	66.7	88.1	76.6	75.6
EuroBERT-2.1B	85.8	73.1	77.1	76.9	73.8	79.0	70.3	79.7	84.2	49.9	88.2	86.5	60.7	63.0	89.9	76.9	75.9

Table 9: Detailed results on multilingual retrieval tasks (nDCG@10, in %).

XNLI																		
Model	European Languages								Extra-European Languages								Average	
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World	
XLM-RoBERTa-280M	80.0	74.8	76.4	75.1	—	—	—	—	71.5	69.1	—	74.4	72.4	74.0	73.2	76.6	74.1	
XLM-RoBERTa-560M	87.1	83.0	83.5	82.9	—	—	—	—	80.4	77.8	—	81.2	80.6	80.2	80.6	84.1	81.7	
XLM-RoBERTa-3.5B	89.0	85.5	85.3	84.5	—	—	—	—	81.5	80.5	—	83.1	82.4	83.3	82.3	86.1	83.7	
mDeBERTa-v3-280M	84.9	81.2	81.1	81.0	—	—	—	—	77.6	76.1	—	79.1	77.7	78.1	78.4	82.0	79.5	
mGTE-MLM-305M	81.1	76.9	78.5	77.2	—	—	—	—	73.6	71.3	—	75.4	72.9	75.9	75.5	78.4	75.8	
ModernBERT-150M	82.8	65.7	69.9	70.6	—	—	—	—	56.9	54.0	—	63.1	53.9	58.3	68.1	72.3	64.3	
ModernBERT-395M	89.4	75.6	79.2	79.1	—	—	—	—	59.6	55.7	—	70.9	60.6	63.0	76.1	80.8	70.9	
EuroBERT-210M	83.5	77.8	79.4	78.9	—	—	—	—	74.3	70.6	—	76.6	74.2	75.1	75.3	79.9	76.6	
EuroBERT-610M	87.8	82.9	84.6	83.6	—	—	—	—	79.5	76.7	—	82.0	80.3	80.8	80.7	84.7	81.9	
EuroBERT-2.1B	89.6	85.5	86.4	85.8	—	—	—	—	82.8	79.9	—	83.3	83.0	82.3	82.3	86.8	84.1	
PAWS-X																		
Model	European Languages								Extra-European Languages								Average	
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World	
XLM-RoBERTa-280M	93.8	86.4	87.5	88.0	—	—	—	—	—	—	—	—	—	—	—	88.9	88.9	
XLM-RoBERTa-560M	95.5	91.0	91.4	91.8	—	—	—	—	—	—	—	—	—	—	—	92.4	92.4	
XLM-RoBERTa-3.5B	95.8	91.9	91.7	92.3	—	—	—	—	—	—	—	—	—	—	—	92.9	92.9	
mDeBERTa-v3-280M	95.7	90.2	90.4	91.3	—	—	—	—	—	—	—	—	—	—	—	91.9	91.9	
mGTE-MLM-305M	94.7	87.5	88.2	88.8	—	—	—	—	—	—	—	—	—	—	—	89.8	89.8	
ModernBERT-150M	94.7	72.0	74.0	77.7	—	—	—	—	—	—	—	—	—	—	—	79.6	79.6	
ModernBERT-395M	95.8	75.4	82.7	83.5	—	—	—	—	—	—	—	—	—	—	—	84.3	84.3	
EuroBERT-210M	95.6	86.5	88.7	88.9	—	—	—	—	—	—	—	—	—	—	—	89.9	89.9	
EuroBERT-610M	95.6	90.0	91.3	92.0	—	—	—	—	—	—	—	—	—	—	—	92.2	92.2	
EuroBERT-2.1B	96.2	91.6	91.8	92.5	—	—	—	—	—	—	—	—	—	—	—	93.0	93.0	
AmazonReviews																		
Model	European Languages								Extra-European Languages								Average	
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World	
XLM-RoBERTa-280M	64.9	65.0	60.6	60.0	—	—	—	—	—	—	59.0	—	—	—	56.8	62.7	61.1	
XLM-RoBERTa-560M	66.9	67.0	62.4	61.5	—	—	—	—	—	—	62.1	—	—	—	58.5	64.5	63.1	
XLM-RoBERTa-3.5B	67.1	67.8	62.4	61.5	—	—	—	—	—	—	63.7	—	—	—	59.2	64.7	63.6	
mDeBERTa-v3-280M	66.4	66.1	61.6	60.6	—	—	—	—	—	—	60.4	—	—	—	57.7	63.7	62.1	
mGTE-MLM-305M	65.0	65.3	60.9	59.5	—	—	—	—	—	—	61.2	—	—	—	57.4	62.7	61.5	
ModernBERT-150M	66.1	61.4	57.5	57.8	—	—	—	—	—	—	54.2	—	—	—	53.9	60.7	58.5	
ModernBERT-395M	67.6	64.9	60.8	60.0	—	—	—	—	—	—	58.2	—	—	—	57.8	63.3	61.5	
EuroBERT-210M	65.9	65.4	60.5	60.2	—	—	—	—	—	—	60.4	—	—	—	57.7	63.0	61.7	
EuroBERT-610M	66.7	66.4	61.6	61.2	—	—	—	—	—	—	61.7	—	—	—	58.1	64.0	62.6	
EuroBERT-2.1B	66.5	67.8	62.8	60.9	—	—	—	—	—	—	62.4	—	—	—	59.0	64.5	63.2	
MassiveIntent																		
Model	European Languages								Extra-European Languages								Average	
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World	
XLM-RoBERTa-280M	89.1	86.1	87.0	86.6	87.5	87.5	86.8	87.3	79.0	86.2	86.5	87.3	85.6	86.7	86.0	87.2	86.3	
XLM-RoBERTa-560M	90.3	87.7	88.0	89.0	88.5	89.1	88.8	88.8	83.5	88.1	88.9	89.0	87.8	88.9	87.1	88.8	88.2	
XLM-RoBERTa-3.5B	89.9	87.6	88.2	88.6	88.7	88.3	87.9	88.6	81.8	88.0	88.4	89.2	87.8	88.4	86.9	88.5	87.9	
mDeBERTa-v3-280M	88.1	86.4	86.9	87.3	87.6	88.0	87.0	86.9	79.8	86.0	87.3	87.3	85.9	86.5	85.9	87.3	86.5	
mGTE-MLM-305M	89.0	86.3	87.4	87.9	87.2	87.9	86.3	87.8	80.7	86.6	87.9	87.8	86.4	87.4	86.3	87.5	86.9	
ModernBERT-150M	85.5	74.8	76.6	79.4	75.1	74.2	71.1	78.2	57.6	56.1	76.6	74.4	62.3	66.3	79.9	76.9	72.6	
ModernBERT-395M	89.8	83.9	84.9	86.9	84.7	84.1	82.4	86.0	72.5	76.7	84.4	83.9	79.6	80.8	84.9	85.3	83.0	
EuroBERT-210M	89.0	86.0	86.9	86.9	87.0	87.1	86.8	87.9	81.2	86.9	87.4	87.2	85.8	85.0	86.0	87.2	86.5	
EuroBERT-610M	89.2	86.6	87.4	87.6	88.1	88.2	87.3	87.8	82.7	87.3	88.3	88.2	86.8	86.1	87.0	87.8	87.2	
EuroBERT-2.1B	88.9	87.2	88.0	88.7	87.9	88.2	88.1	88.2	83.2	87.6	89.0	88.1	87.1	85.4	87.0	88.2	87.5	

Table 10: Detailed results on multilingual sequence classification tasks (accuracy, in %).

Model	Ref-free															
	European Pairs						Extra-European Pairs								Average	
	en-xx		xx-en		Other		en-xx				xx-en				Euro	World
	en-de	en-pl	de-en	pl-en	de-fr	fr-de	en-ja	en-ru	en-tr	en-zh	ja-en	ru-en	tr-en	zh-en		
XLM-RoBERTa-280M	45.3	53.6	26.7	16.3	31.4	32.0	47.0	56.5	61.5	42.5	10.4	21.8	40.7	25.3	34.2	36.5
XLM-RoBERTa-560M	50.7	66.0	30.7	15.8	41.1	29.8	52.1	61.2	66.2	47.2	11.0	24.8	45.4	29.0	39.0	40.8
XLM-RoBERTa-3.5B	55.1	66.9	35.9	18.7	46.7	43.3	56.3	64.9	64.8	52.2	13.1	27.0	47.7	30.6	44.4	44.5
mDeBERTa-v3-280M	52.8	61.6	33.1	18.3	44.4	39.6	51.6	61.2	67.2	46.7	11.2	24.2	47.3	28.9	41.6	42.0
mGTE-MLM-305M	48.6	55.2	30.4	18.5	37.5	35.9	48.3	57.2	59.7	45.5	10.6	23.4	41.5	27.3	37.7	38.5
ModernBERT-150M	39.7	47.4	29.8	18.2	20.5	21.9	36.0	41.6	39.6	37.8	10.7	21.5	41.3	23.9	29.6	30.7
ModernBERT-395M	45.3	51.7	32.4	20.0	23.8	27.6	37.8	44.7	43.6	41.5	12.3	24.1	42.7	27.1	33.5	33.9
EuroBERT-210M	52.9	58.4	33.2	17.5	40.6	40.3	51.1	57.9	57.3	48.3	14.3	26.7	44.3	30.8	40.5	41.0
EuroBERT-610M	52.9	61.1	32.4	18.2	42.6	39.2	51.3	59.4	62.3	48.6	12.2	26.6	44.1	29.7	41.1	41.5
EuroBERT-2.1B	49.1	57.8	29.8	19.3	38.3	38.5	47.8	56.9	56.5	45.0	10.7	23.5	41.3	27.5	38.8	38.7

Model	Ref-based															
	European Pairs						Extra-European Pairs								Average	
	en-xx		xx-en		Other		en-xx				xx-en				Euro	World
	en-de	en-pl	de-en	pl-en	de-fr	fr-de	en-ja	en-ru	en-tr	en-zh	ja-en	ru-en	tr-en	zh-en		
XLM-RoBERTa-280M	49.6	56.0	34.4	29.2	47.8	41.9	49.8	60.5	63.1	50.1	14.2	25.3	48.7	31.1	43.1	43.0
XLM-RoBERTa-560M	52.3	63.9	37.3	26.8	51.1	42.3	53.6	63.3	70.9	53.0	12.6	25.2	49.2	32.3	45.6	45.3
XLM-RoBERTa-3.5B	55.9	68.2	39.1	28.2	53.7	45.6	56.7	66.3	71.1	55.2	12.8	28.9	52.4	34.0	48.5	47.7
mDeBERTa-v3-280M	53.0	65.6	37.6	27.9	50.9	43.7	52.7	63.5	69.1	53.0	13.1	27.8	49.9	32.5	46.5	45.7
mGTE-MLM-305M	51.0	57.4	36.6	29.9	49.6	39.3	51.5	61.8	63.5	52.4	13.8	26.0	48.7	32.8	44.0	43.9
ModernBERT-150M	43.5	50.2	36.8	28.2	42.5	33.1	45.4	50.4	54.2	46.0	13.8	26.6	50.8	31.7	39.1	39.5
ModernBERT-395M	47.4	53.9	39.2	31.6	43.5	34.8	47.1	52.3	60.0	48.6	14.9	28.9	53.1	34.0	41.7	42.1
EuroBERT-210M	52.5	59.8	38.8	30.5	49.7	39.6	52.2	61.5	65.9	53.1	15.2	28.7	50.5	34.7	45.1	45.2
EuroBERT-610M	53.8	64.0	37.8	29.4	51.3	42.5	53.2	62.9	67.9	52.3	16.2	29.2	50.6	33.5	46.5	46.0
EuroBERT-2.1B	54.9	65.7	39.4	31.0	54.9	45.0	55.0	64.3	68.3	54.5	14.9	28.3	52.0	34.3	48.5	47.3

Table 11: Detailed results on the WMT tasks (Spearman rank correlation, in %).

Model	European Languages								Extra-European Languages							Average	
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World
XLM-RoBERTa-280M	53.2	55.6	62.0	—	—	—	—	—	—	—	—	69.1	67.3	59.3	—	56.9	61.1
XLM-RoBERTa-560M	57.1	60.1	66.9	—	—	—	—	—	—	—	—	73.6	72.6	62.7	—	61.4	65.5
XLM-RoBERTa-3.5B	58.1	62.1	69.7	—	—	—	—	—	—	—	—	75.6	75.5	64.0	—	63.3	67.5
mDeBERTa-v3-280M	56.2	58.5	66.3	—	—	—	—	—	—	—	—	72.6	71.9	59.6	—	60.3	64.2
mGTE-MLM-305M	52.7	58.6	66.3	—	—	—	—	—	—	—	—	69.7	69.6	61.4	—	59.2	63.0
ModernBERT-150M	44.6	47.9	52.8	—	—	—	—	—	—	—	—	59.6	53.8	52.1	—	48.4	51.8
ModernBERT-395M	56.9	56.8	64.0	—	—	—	—	—	—	—	—	66.7	65.6	59.0	—	59.3	61.5
EuroBERT-210M	54.4	58.7	67.2	—	—	—	—	—	—	—	—	70.0	71.2	61.3	—	60.1	63.8
EuroBERT-610M	57.2	60.6	70.3	—	—	—	—	—	—	—	—	72.6	73.5	61.5	—	62.7	66.0
EuroBERT-2.1B	58.8	62.1	71.1	—	—	—	—	—	—	—	—	74.4	75.7	62.8	—	64.0	67.5

Table 12: Detailed results on the SeaHorse summary evaluation task (Spearman rank correlation, in %).

Model	European Languages								Extra-European Languages							Average	
	en	de	es	fr	it	nl	pl	pt	ar	hi	ja	ru	tr	vi	zh	Euro	World
XLM-RoBERTa-280M	97.6	94.5	93.6	—	—	96.1	—	—	—	—	—	—	—	—	—	95.5	95.5
XLM-RoBERTa-560M	97.7	96.4	93.9	—	—	96.2	—	—	—	—	—	—	—	—	—	96.1	96.1
XLM-RoBERTa-3.5B	98.3	96.4	94.5	—	—	96.1	—	—	—	—	—	—	—	—	—	96.3	96.3
mDeBERTa-v3-280M	98.1	96.4	94.2	—	—	96.1	—	—	—	—	—	—	—	—	—	96.2	96.2
mGTE-MLM-305M	97.9	94.0	93.0	—	—	95.7	—	—	—	—	—	—	—	—	—	95.2	95.2
ModernBERT-150M	84.0	89.4	89.8	—	—	92.8	—	—	—	—	—	—	—	—	—	89.0	89.0
ModernBERT-395M	97.8	81.5	92.5	—	—	94.6	—	—	—	—	—	—	—	—	—	91.6	91.6
EuroBERT-210M	97.7	91.8	93.8	—	—	95.5	—	—	—	—	—	—	—	—	—	94.7	94.7
EuroBERT-610M	97.6	96.0	94.0	—	—	95.8	—	—	—	—	—	—	—	—	—	95.9	95.9
EuroBERT-2.1B	97.6	94.2	93.8	—	—	95.0	—	—	—	—	—	—	—	—	—	95.2	95.2

Table 13: Detailed results on the NER task (F1 score, in %).